

JavaScript Async/Await

Async / Await

This is the concept that makes JavaScript handle real-world tasks like:

- ✓ API calls
- ✓ Database operations
- ✓ Playwright automation
- ✓ waiting for elements
- ✓ handling dynamic web pages

Without async/await, automation becomes flaky and unreliable.

What is Async/Await in JavaScript?

JavaScript normally runs code line by line.

But in real life, many tasks take time, like:

- loading a webpage
- fetching API response
- waiting for a button to appear

So JavaScript provides async/await to handle such operations smoothly.

In simple words:

- ✓ async means: "This function will take time."
 - ✓ await means: "Wait here until the work is completed."
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Real-Life Example

Imagine you order food :

- You place an order
- You must wait for delivery
- Only after delivery you can eat

If you don't wait, you'll start eating without food

That's exactly what happens in code without await.

Example Without Await (Problem)

```
console.log("Start");
```

```
fetch("https://api.com/data");
```

```
console.log("End");
```

Output:

Start

End

(API response comes later)

 The program does not wait.

Example With Async/Await (Correct Way)

```
async function getData() {
```

```
  console.log("Start");
```

```
  let response = await fetch("https://api.com/data");
```

```
console.log("Data received");  
}
```

```
getData();
```

 Now JavaScript waits properly.

Real Usage in Automation / Playwright

Playwright is incomplete without async/await.

Example:

```
await page.goto("https://example.com");
```

```
await page.fill("#username", "admin");
```

```
await page.fill("#password", "secret");
```

```
await page.click("#loginBtn");
```

 Here every action takes time.
So Playwright uses await to avoid flaky tests.

What's Covered in This Chapter?

-  Why async/await is needed
-  Difference between normal code and async code
-  How await prevents flaky automation
-  Real Playwright examples
-  Interview-focused understanding

Key Learning Points

- ◆ **async functions return promises**
 - ◆ **await pauses execution until task completes**
 - ◆ **Used in API testing and automation**
 - ◆ **Helps avoid timing issues in scripts**
 - ◆ **Top interview topic for JavaScript + SDET**
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Why This Matters in Interviews?

Interviewers check:

- 📌 **understanding of asynchronous behavior**
- 📌 **real-world coding mindset**
- 📌 **ability to handle API and automation flows**
- 📌 **knowledge of promises and await usage**

 **Strong async/await = strong JavaScript engineer.**

“Automation becomes stable when your code learns to wait smartly.”
